Famer Jones example

Friday, 20 February 2026 12:31 PM

chocolate	\$4	20mins	250mL	4 eggs
plain	\$2	50mins	200mL	1 egg

constraints

8 hours = 480mins

20 eggs

5L = 5000mL

chocolate = c

plain = p

revenue = $4c + 2p$

480mins = $20c + 50p$

20eggs = $4c + p$ matrix?

5000mL = $250c + 200p$

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lecturer's notes

Sets

C cakes $\{\overset{1}{\text{choc}}, \overset{2}{\text{plain}}\}$

I ingredients $\{\text{milk}, \text{eggs}, \text{time}\}$

Data

r_c revenue (\$) for cake $c \in C$

a_i availability of ingredient $i \in I$

u_{c_i} usage of ingredient $i = I$ in cake $c \in C$

Variable

x_c amount of cake $c \in C$ to bake

objective

maximise revenue

objective

maximize revenue

$z = 4x_1 + 2x_2$

constraints

$30 \geq 4x_1 + 1x_2$ (eggs)

$480 \geq 20x_1 + 50x_2$ (time)

$5000 \geq 250x_1 + 200x_2$ (milk)

$0 \leq x_1, x_2$ (non-negative)

$x_1, x_2 \in \mathbb{Z}$

Linear Programming Problem (LP)

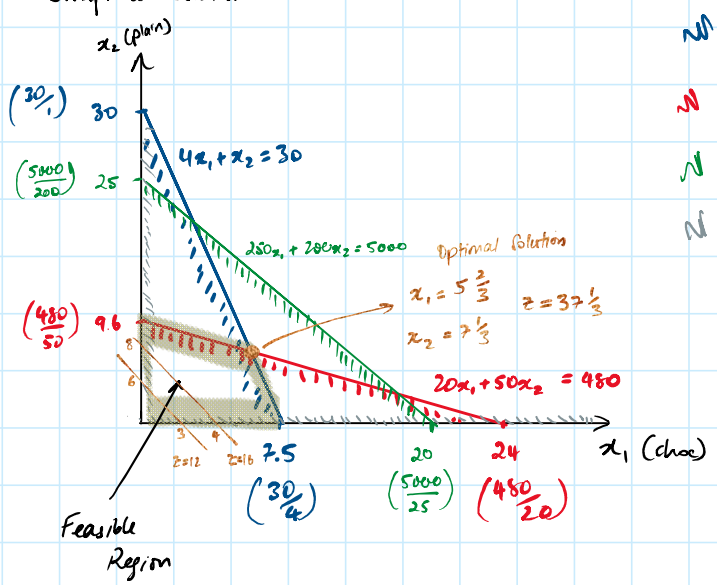
Integer Programming Problem (IP)

Equations:

objective: $\max z = \sum_{c \in C} r_c x_c$

constraints $\sum_{c \in C} u_{ci} x_c \leq a_i \quad \forall i \in I$

Graphical Solution



eggs

time

milk

other (non negative, int)

↳ find which point in feasible region makes highest revenue

$z = 4x_1 + 2x_2$

$z = 12$ x_1 intercept 3

x_2 intercept 6 → parallel lines (see graph)

$z = 12$ x_1 intercept 3
x_2 intercept 6 $\longrightarrow$

parallel lines (see graph)

$z = 16$ x_1 intercept 4
x_2 intercept 8 $\longrightarrow$

- maximum point within constraints